

Rodrigo Díaz Velasco

Senior AI Engineer & Technical Leader | GenAI, LLM & Agentic AI

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Summary

Senior AI Engineer and Technical Leader with over 20 years of combined experience in technology project management, entrepreneurship, and AI engineering. Designs and deploys Generative AI (GenAI) solutions to production: Large Language Model (LLM) agents built with OpenAI, LangGraph, and LangChain, Retrieval-Augmented Generation (RAG) with embeddings and vector databases, and prompt engineering (zero-shot, few-shot, chain-of-thought). Experienced in leading multidisciplinary teams of up to 10 people, executive reporting, and turning business needs into production-ready solutions across international, cross-cultural environments. Active tech community speaker.

Skills

- **GenAI & LLMs:** OpenAI GPT, Anthropic Claude, LLaMA, Prompt Engineering, RAG, Embeddings, Agentic AI, LangGraph, LangChain, Guardrails, Model Evaluation & Selection (cost, performance, quality), Natural Language Processing (NLP).
- **Software Engineering & MLOps:** Python, FastAPI, Docker, Git, CI/CD, Automated Testing, MLOps, LangSmith & Sentry (Observability), Weights & Biases, Cloud Deployment (Render, Vercel, DigitalOcean).
- **Data & Retrieval:** Pinecone (Vector DB), Vector Search (text and image), Neo4j (Graph DB), Machine Learning, NumPy, Pandas, SQL, R.
- **Leadership & Management:** Technical Leadership, Cross-functional Team Leadership, Stakeholder Management, Executive Reporting, KPI & Risk Management, Business Case & ROI Validation.
- **Languages:** English (C1; bilingual education, FCE), Spanish (Native).

Professional Experience

Senior AI Engineer / Technical Lead

Independent Consultant | Feb 2021 – Present

- **End-to-End GenAI Delivery:** Lead the full lifecycle of AI solutions for clients in Argentina and Mexico, from problem definition to production deployment of FastAPI backends.
- **Agentic AI & RAG:** Build LLM-based agents using OpenAI models orchestrated with LangGraph and LangChain, combining RAG with embeddings, Pinecone vector and Neo4j graph databases.
- **Prompt Engineering:** Design and optimize prompts using zero-shot, few-shot, and chain-of-thought strategies, with reusable prompt templates for multi-prompt pipelines.
- **Engineering Best Practices:** Structure every project with Git, CI/CD, automated testing, guardrails, and observability (LangSmith, Sentry alerts); compare LLMs on cost, performance, and quality to select the right model.
- **Featured Project (Smart Search):** Directed a hybrid intent-driven search engine with a multi-prompt pipeline on OpenAI models and LangGraph, using embeddings in a dual-index Pinecone strategy to maximize precision and minimize hallucinations. (**Link:** theviderlab.github.io/smart-search-project)
- **Direct Booking Engine & LLM Assistant (Apartur):** Planned and developed, together with the client's CTO, a direct booking engine for a 4-hotel chain in Argentina that increased direct bookings by over 50%, with a REST API for an LLM-based virtual assistant. (**Link:** apartur.com.ar)

Co-founder & Managing Partner

Invagrup (FoodTech Startup) | Oct 2013 – Jan 2021

- **Business Strategy & Ownership:** Led business development, strategy, and operations with full accountability for results.
- **Executive Collaboration:** Worked directly with CEOs, business partners, and clients to drive growth and make strategic decisions on market positioning.
- **Business Case Validation:** Identified business opportunities and validated their viability and return on investment.
- **Industry Recognition:** Finalist in the NAVES entrepreneurship competition by IAE Business School (2019); recognized at the APSAL Awards (2020) as an outstanding food-industry innovation proposal.

Project Manager

Schneider Electric | Mar 2012 – Oct 2013

- **Multidisciplinary Leadership:** Led engineering initiatives across different countries, adapting management and collaboration styles to diverse cultural environments.
- **Delivery & Reporting:** Monitored progress, risks, and KPIs, and presented performance reports to senior management and executive stakeholders.

Project Manager

ABB | Feb 2011 – Feb 2012

Lead Project Engineer

Micro Automation | Oct 2007 – Jan 2011

Project Engineer

Laboratorio Argentino de Cine (LACSA) | Jan 2007 – Sep 2007

Project Manager

Telefónica | Jan 2005 – Dec 2006

Education & Certifications

- **Master's Degree in Artificial Intelligence** | Universidad Internacional de Valencia (VIU) (Oct 2023 – Sep 2025) — GPA: 9.28/10 (Three Honors Distinctions); includes NLP
- **Electronic Engineering Degree (Mechatronics)** | Instituto Tecnológico de Buenos Aires (ITBA) (Mar 2000 – Sep 2006)
- **Postgraduate in Project Management (PMI)** | ITBA (Jan 2006 – Nov 2006)
- **MLOps Certification** | Universidad de Extremadura (2026) — End-to-end lifecycle management
- **Fundamentals of Accelerated Computing with CUDA C/C++** | NVIDIA (2024)
- **Data Science Training** | DataCamp (2021 – 2025) — R, Python, scikit-learn, Pandas, NumPy

Volunteer Experience

Technology Speaker & Collaborator

Extremadura Tech Community | Ongoing

- Deliver public talks on AI and emerging technologies for the regional tech ecosystem.
- Key collaborator for Extremadura Digital Day, the region's annual technology event, communicating technical ideas clearly to foster technology adoption.